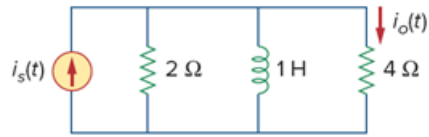


Problem

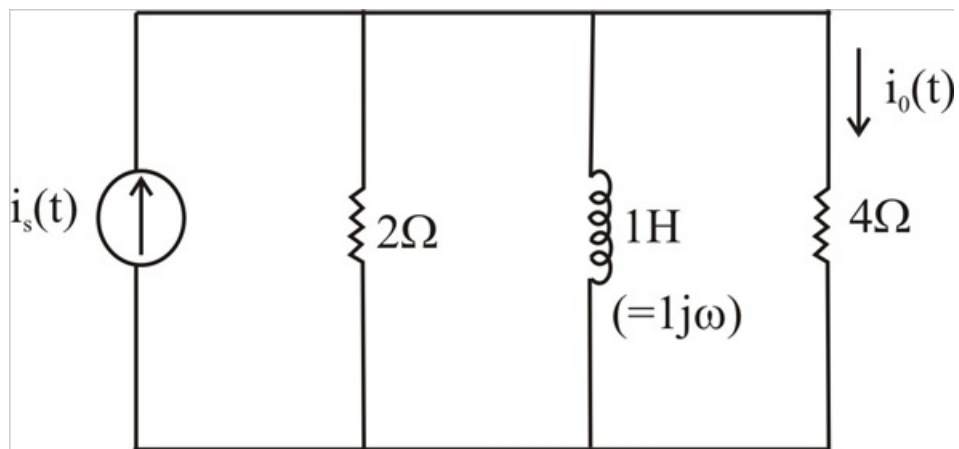
Find the transfer function $I_o(\omega)/I_s(\omega)$ for the circuit in Fig. 18.39.

Figure 18.39



Step-by-step solution

Step 1 of 2



Step 2 of 2

$$\begin{aligned}
 \frac{I_o(\omega)}{I_s(\omega)} &= \frac{2j\omega}{2j\omega + 8 + 4j\omega} \left[\frac{Z_1 Z_2}{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1} \right] \\
 &= \frac{2j\omega}{8 + 6j\omega} \\
 &= \boxed{\frac{j\omega}{4 + 3j\omega}}
 \end{aligned}$$